GEOMETRY · SIMILARITY

Dilation & Similarity Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Dilation & Similarity Lab** on Mr. K's Classroom: Geometry → Similarity → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. If every side of a triangle doubles, what happens to its area?

   It becomes 4 times as large.

2 Explore use the tool and record what you see

Dilate △ABC with A(1, 1), B(4, 1), C(1, 3)

Original: AB = 3, perimeter ≈ 8.61, area = 3. Change k and record.

k	A′B′	Perimeter ratio	Area ratio	Image area
2	**6**	**2**	**4**	**12**
3	**9**	**3**	**9**	**27**
$\frac{1}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
$\frac{3}{2}$	$\frac{9}{2}$	$\frac{3}{2}$	$\frac{9}{4}$	$\frac{27}{4}$

What stays the same?

	Preimage	Image (k = 2)
Angles	**90°, 33.7°, 56.3°**	**90°, 33.7°, 56.3°**
Side ratios AB : AC	**3 : 2**	**6 : 4 = 3 : 2**

3 Write the rule fill in each blank from what you observed

1. A dilation multiplies every length by **k**.
2. Angles in the image are **equal to** the angles in the preimage.
3. Perimeter is multiplied by **k** and area by **k^2**.
4. $k > 1$ makes an **enlargement**. $0 < k < 1$ makes a **reduction**.
5. The center of dilation is the one point that **doesn't move**.

4 Check no tool: show your work

1. A segment of length 2.5 is dilated by k = 4. How long is the image?

   10

2. Similar triangles have corresponding sides 6 and 9. Find the scale factor from small to large.

   $\frac{3}{2}$

3. A figure with area 20 is dilated by k = 3. Find the new area.

   180

4. Dilate (2, −4) by $\frac{1}{2}$ about the origin.

   (1, −2)

5 Reflect

Why does the area grow by k^2 instead of k?

Area uses two dimensions, and both length and width are multiplied by k.